

| Model                           | Dev         | Test         |
|---------------------------------|-------------|--------------|
| XLNet + RoBERTa (LDL)           | 0.547       | 0.518        |
| XLNet + BiLSTM-ELMo (Keyphrase) | 0.538       | 0.532        |
| XLNet + BiLSTM-ELMo (LDL)       | <b>0.55</b> | <b>0.543</b> |

Table 7: Performance of different ensemble models

sponds to a particular sentence belonging to a presentation slide in the original corpus. The development set results can be found in Table 8. The evaluation scheme used in this experiment uses the same  $Match_m$  as described in the Evaluation Metric section but with  $m = 1, 2, 3, 4$  as used in ?.

| Model             | Dev          |
|-------------------|--------------|
| XLNet             | <b>0.758</b> |
| XLNet (LDL)       | 0.757        |
| RoBERTa           | 0.743        |
| RoBERTa (LDL)     | 0.745        |
| BiLSTM-ELMo       | 0.751        |
| BiLSTM-ELMo (LDL) | 0.752        |

Table 8: Sentence-wise results on the Development set

- i) It is extremely important that parents take time to SLOW DOWN and give their child their undivided attention. The importance of that can not be over-emphasized.
- ii) It is extremely important that parents take time to SLOW DOWN and give their child their undivided attention. The importance of that can not be over-emphasized.
- iii) It is extremely important that parents take time to SLOW DOWN and give their child their undivided attention. The importance of that can not be over-emphasized.
- iv) It is extremely important that parents take time to SLOW DOWN and give their child their undivided attention. The importance of that can not be over-emphasized.

Figure 5: Emphasis Heatmaps i) Ground Truth ii) BiLSTM-ELMo iii) XLNet iv) Best Ensemble Model

## Analysis

### Length vs Performance

We wanted to understand how the performance of our models was affected by the length of the instances. Table 9 summarizes the performance of our best performing single model, i.e., XLNet on the development set divided into three sets, Short ( $\leq 40$  tokens, 80 samples), Medium (40 to 90 tokens, 262 samples), and Long ( $>90$  tokens, 50 samples). As we can see, the model performance deteriorates with the increasing length of the instances.

|                                | XLNet        |
|--------------------------------|--------------|
| Small ( $\leq 40$ )            | <b>0.648</b> |
| Medium ( $>40$ and $\leq 90$ ) | 0.549        |
| Large ( $>90$ )                | 0.42         |

Table 9: Average  $Match_m$  for best performing XLNet model on different size of instances in the development set

## Emphasis vs Parts of Speech

Table 10 shows POS (Parts of Speech) tags vs. average emphasis on the development dataset. We did this experiment to understand how our model predictions performed on each POS tag when compared to the actual human-annotated emphasis scores on the development set. We noticed that the original average emphasis scores were highest on Adjectives followed by Noun. On comparing our models, we found that XLNet was able to almost accurately predict the emphasis scores on Adjectives and Noun respectively, and BiLSTM-ELMo also had the highest predictions on Adjectives and Noun respectively. We also noticed that XLNet did a better job on predicting the emphasis score on different POS tags where the predictions were either very close to the human scores or marginally lesser. On the other hand, we noticed that BiLSTM-ELMo’s predictions fell short by bigger margins when compared to XLNet and gave more emphasis to Adverbs than that in the development set.

| POS        | Count | Human | BiLSTM | XLNet |
|------------|-------|-------|--------|-------|
| Noun       | 4719  | 0.169 | 0.134  | 0.168 |
| Verb       | 1420  | 0.118 | 0.083  | 0.113 |
| Adjectives | 982   | 0.186 | 0.140  | 0.181 |
| Det        | 634   | 0.062 | 0.029  | 0.042 |
| Adverbs    | 347   | 0.111 | 0.068  | 0.103 |
| Pronouns   | 165   | 0.040 | 0.068  | 0.022 |
| Punct      | 2082  | 0.034 | 0.015  | 0.025 |

Table 10: POS tags vs. average emphasis on development dataset

## Conclusion

In this paper, we present our approach to AAAI-CAD21 shared task: Predicting Emphasis in Presentation Slides. Our best submission gave us an average  $Match_m$  of 0.518 placing us 3<sup>rd</sup> on the Evaluation phase leaderboard and an average  $Match_m$  of 0.543 placing us 1<sup>st</sup> on the Post-Evaluation leaderboard at the time of writing the paper. Future work includes using a hierarchical approach to emphasis prediction as a sequence labeling task using both sentence-level (individual sentence in a slide) and slide-level representations of a word (?).

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